

Dynamic proportion portfolio insurance using genetic programming with principal component analysis

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Abstract

This paper proposes a **dynamic proportion portfolio insurance (DPPI)** strategy based on the popular **constant proportion portfolio insurance (CPPI)** strategy. The constant multiplier in CPPI is generally regarded as the risk multiplier. Since the market changes constantly, we think that the risk multiplier should change according to market conditions. This research identifies risk variables relating to market conditions. These risk variables are used to build the equation tree for the risk multiplier by genetic programming. Experimental results show that our DPPI strategy is more profitable than traditional CPPI strategy. In addition, principal component analysis of the risk variables in equation trees indicates that among all the risk variables, risk-free interest rate influences the risk multiplier most.

Constant proportion portfolio insurance(CPPI)

- A simply strategy based on the investor's preference of risk to calculate the amount invested in the risky asset.(Black & Jones, 1987).

Formula : $E = M * (A - F)$

Where

- Exposure E is the dollar amount invested in the active(risky) asset
- Asset A is the value of portfolio
- Floor F is the lowest acceptable value of the portfolio
- $(A-F)$ is the cushion as the excess of the portfolio value over the floor
- Risk multiplier M is the ratio of the initial exposure of the initial cushion

Note :

The higher risk multiplier, the higher risk the investor take.

CPPI Example

假設小王於股市的投資組合總值為 300 萬元 (A) ， CPPI 乘數 (M) 為 3 ， 保險額度 (A) 為 240 萬元。

所以配置於股票 (風險性資產) 的額度 = $3 * (300 - 240) = 180$ 萬元

無風險資產的額度為 $= (300 - 180) = 120$ 萬元

Case1 : 股價下跌

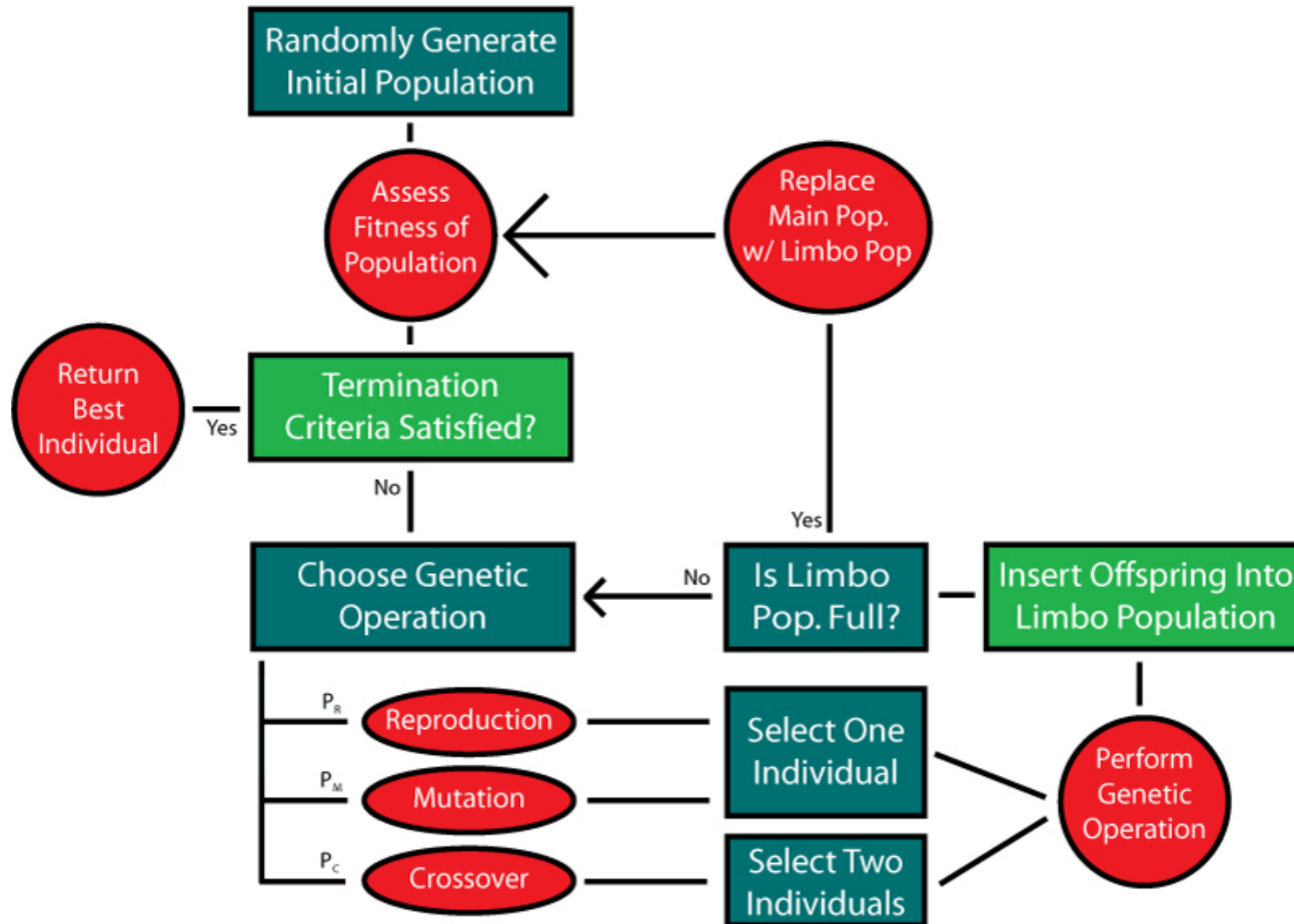
日期	組合總市值 (A)	風險性資產 (E)	無風險資產	股票加碼
1/25	300	180	120	
1/26	280	120	160	-40
1/27	270	90	180	-20
1/28	240	0	240	-60

Case2 : 股價上漲

日期	組合總市值 (A)	風險性資產 (E)	無風險資產	股票加碼
1/29	300	180	120	
1/30	330	270	60	60
1/31	360	360	20	60
2/1	380	420	-40(融資)	40

Genetic programming(GP)

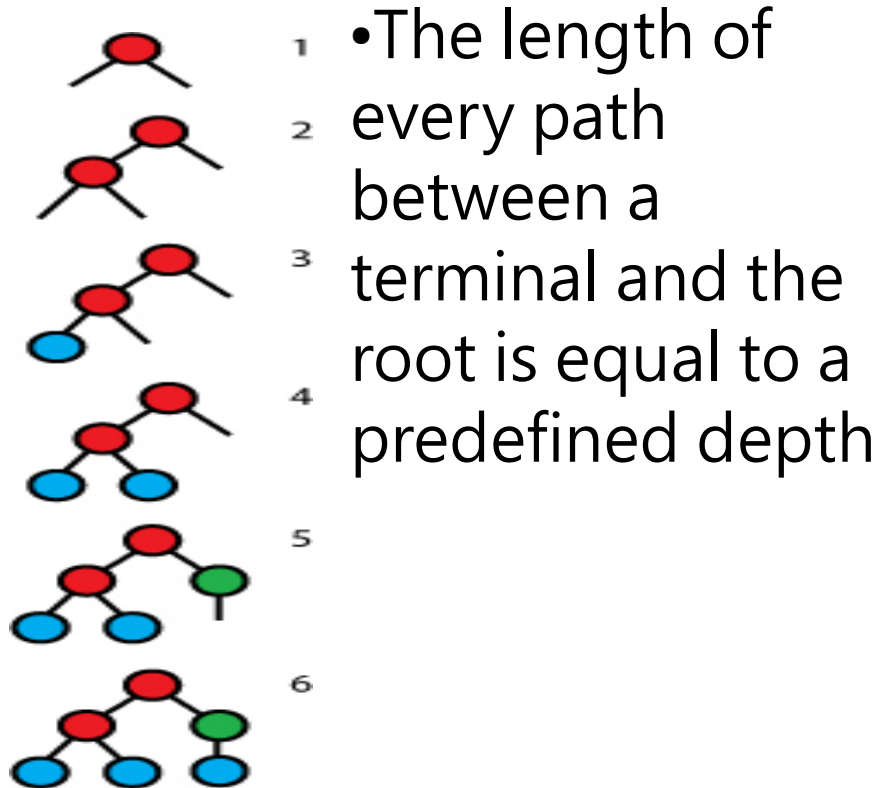
- Genetic programming is a branch of genetic algorithms.
- The [main difference](#) between genetic programming and genetic algorithms is [the representation of the solution](#).



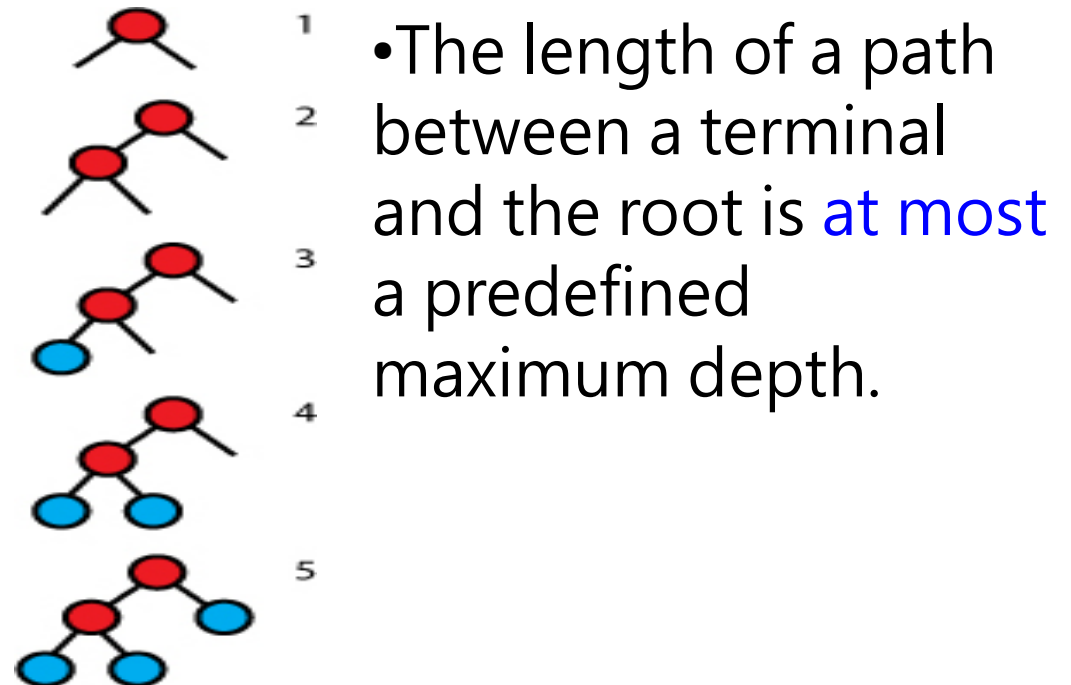
Initial population

- We have determined the **MAX DEPTH**

1 、 Full method

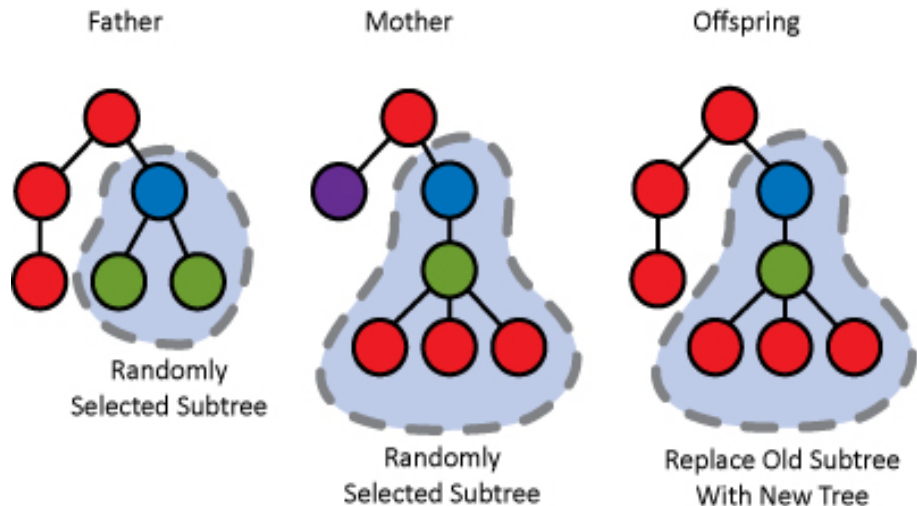


2 、 Grow method



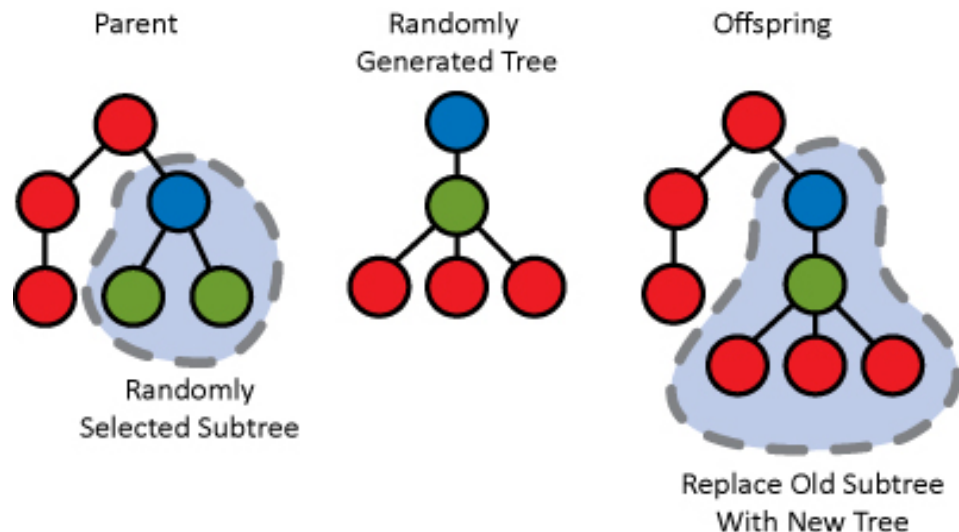
Usually, an equal mix of trees is produced with the Full and Grow methods.

Crossover(Subtree crossover)



The probability distribution for the selection of a crossover point favors the exchange of subtrees rather than terminals.

Mutation(Subtree mutation)



- First, a random point is chosen in the tree.
- This point, as well as the subtree below it, are then removed and replaced by a new, randomly generated subtree.

Principal component analysis(PCA)

A way of identifying patterns in data.

Step :

- 1 、 get data set
- 2 、 subtract the mean
- 3 、 calculate the covariance matrix
- 4 、 calculate the eigenvectors and eigenvalues of the covariance matrix
- 5 、 Choosing components and forming a feature vector

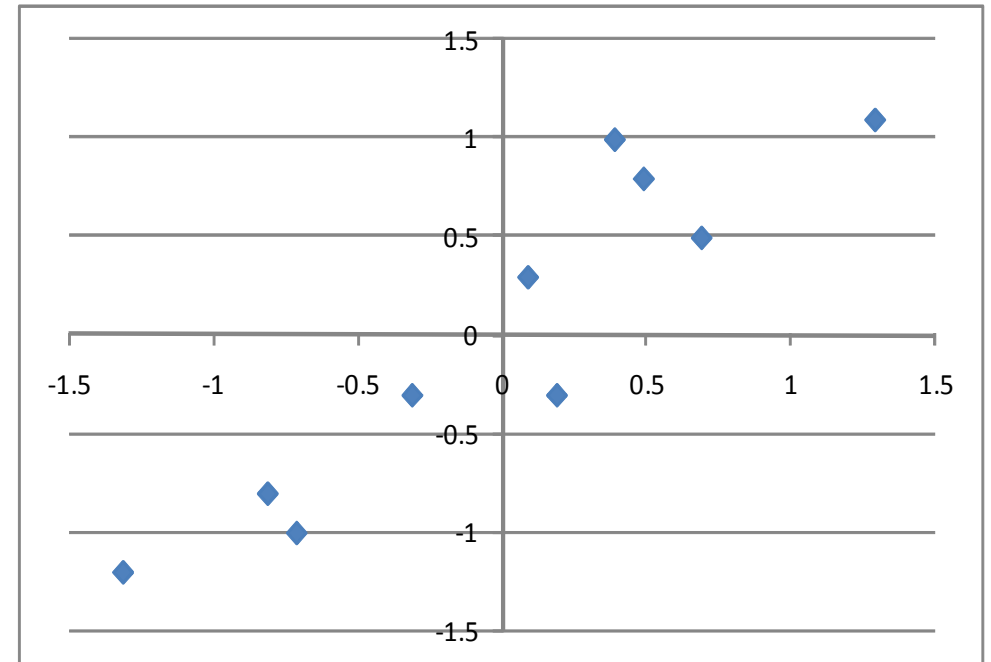
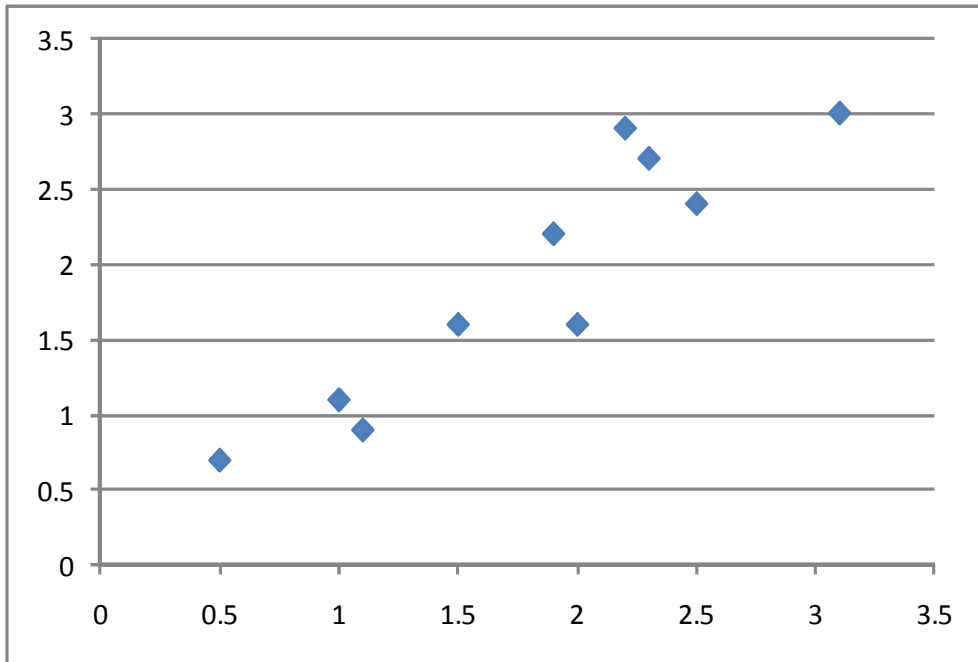
Step 1, 2 :

	x	y
Data =	2.5	2.4
	0.5	0.7
	2.2	2.9
	1.9	2.2
	3.1	3.0
	2.3	2.7
	2	1.6
	1	1.1
	1.5	1.6
	1.1	0.9

$\text{avg}(x) = 1.81$
 $\text{avg}(y) = 1.91$

	x	y
DataAdjust =	.69	.49
	-1.31	-1.21
	.39	.99
	.09	.29
	1.29	1.09
	.49	.79
	.19	-.31
	-.81	-.81
	-.31	-.31
	-.71	-1.01

$\text{avg}(x) = 0$
 $\text{avg}(y) = 0$



Step 3 : calculate the covariance matrix

$$\text{cov}(x, y) = \begin{pmatrix} 0.61655556, & 0.61544444 \\ 0.61544444, & 0.71655556 \end{pmatrix}$$

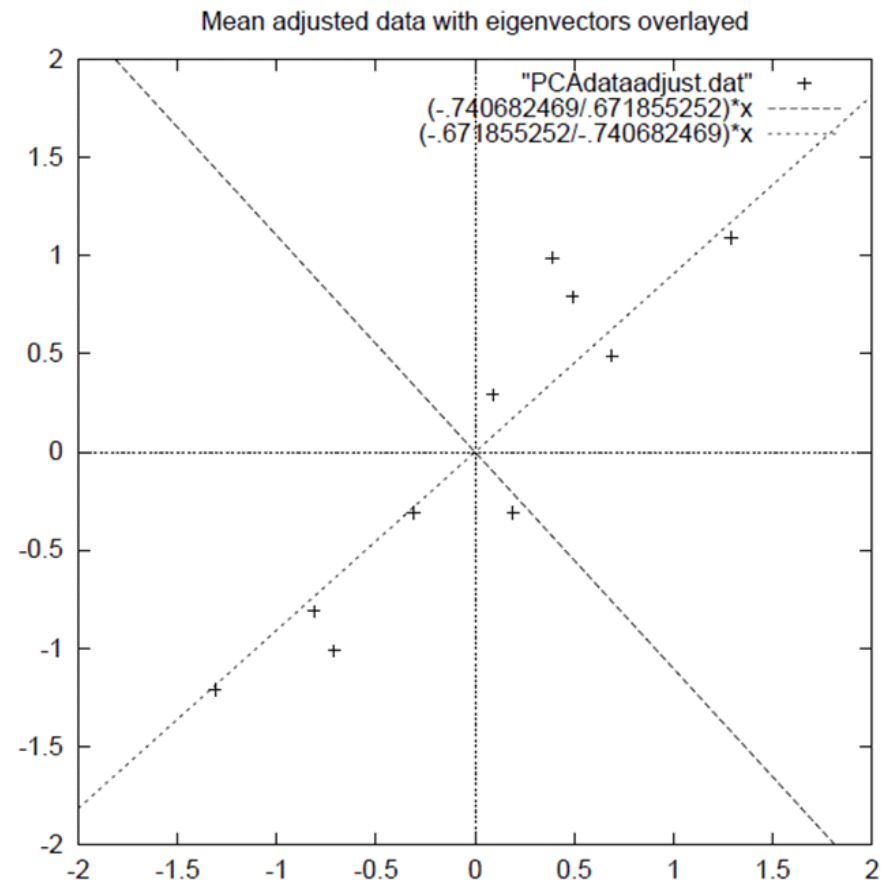
Step 4 : calculate the eigenvectors and eigenvalues of the covariance matrix :

Eigenvalue = ([0.0490834 , 1.28402771])

Eigenvector = [-0.73517866, 0.6778734]
[-0.6778734 , -0.73517866]

Order the eigenvector by eigenvalue,
highest to lowest(the order of significance)

Eigenvector (new order)=
[-0.6778734, -0.73517866]
[-0.73517866, 0.6778734]



Step 5 : Choosing components and forming a feature vector

Single eigenvector :

$$\begin{bmatrix} -0.6778734, & -0.73517866 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix}$$

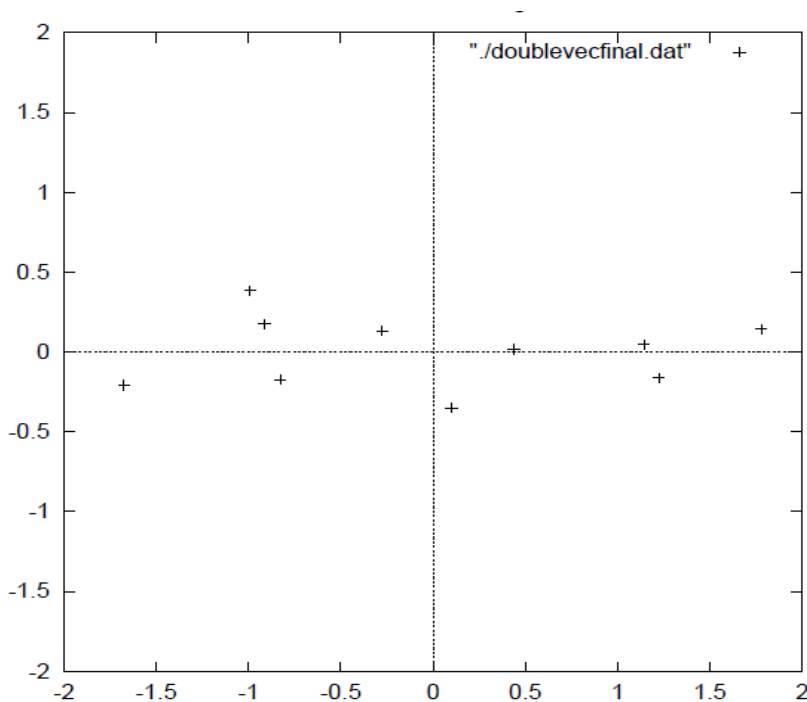
2 eigenvector :

$$\begin{bmatrix} -0.6778734, & -0.73517866 \\ -0.73517866, & 0.6778734 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix}$$

Transformed Data (Single eigenvector)

x
-.827970186
1.77758033
-.992197494
-.274210416
-1.67580142
-.912949103
.0991094375
1.14457216
.438046137
1.22382056

x	y
-.827970186	-.175115307
1.77758033	.142857227
-.992197494	.384374989
-.274210416	.130417207
-1.67580142	-.209498461
-.912949103	.175282444
.0991094375	-.349824698
1.14457216	.0464172582
.438046137	.0177646297
1.22382056	-.162675287



Dynamic proportion portfolio insurance(DPPI)

- The risk multiplier in DPPI becomes a function of risk variables(constructed by GP) instead of a constant as in CPPI.

Formula : $E = M(\gamma) * (A - F)$

The terminal set :

- 1 、 constants
- 2 、 risk variables(market volatility factors and technical indicators)

The function set :

+(addition), -(subtraction), x(multiplication), /(division)

Market volatility factors :

- Risk-free rate(T-bill, 30-day)
- Market index(NYSE Composite)
- Exchange rate(US dollar to Euro)
- Trading volume of the whole market
- Trading volume of the individual stock

Technical indicator :

- Disparity index(DI)
- Relative strength index(RSI)
- Stochastic oscillator(STO)
- Williams %R(%R)
- Money flow index(MFI)

Fitness function :

$$f = \frac{r + 1}{\sigma + 1}$$

r : rate of return after the transaction cost and tax are deduced.

σ : the standard deviation of r

To avoid unnecessary transaction costs due to frequent minor adjustments, the exposure is readjusted when the difference between the current exposure and exposure calculated by the DPPI formula is greater than 5%.

To determine the contributions of the risk variables to the risk multiplier, principal component analysis(PCA) is used to analyze their frequencies in the equation trees generated by genetic programming.

Experiment

Table 4

The five companies used in this study

Symbol	Company	Industry
IBM	Int. Business machines	Computer hardware
LEE	Lee enterprises	Printing and publishing
MOT	Motorola	Comm. equipment
UPS	United parcel service	Trucking
XOM	Exxon mobil	Oil and Gas

Table 5

Training and testing periods of the experiments

Training period	Testing period
2001.07–12	2002.01–06
2002.01–06	2002.07–12
2002.07–12	2003.01–06
2003.01–06	2003.07–12
2003.07–12	2004.01–06
2004.01–06	2004.06–12

Table 6

Parameters of our genetic programming runs

Parameter	Value
Population size	500
Termination condition	200 generations
Maximum tree depth	8
Selection method	Roulette wheel
Crossover method	Subtree crossover
Crossover rate	100% $\searrow$ 15% Decreasing with the generation
Mutation method	Subtree replacement
Mutation rate	1%
Elitism	Top two of a generation

Table 7
Performances of CPPI and DPPI on testing periods

Period	Stock	CPPI		DPPI	
		Fitness	Adj.	Fitness	Adj.
2002.01–06	IBM	0.8608	111	0.9257	33
	LEE	0.8011	29	0.9470	35
	MOT	0.8647	88	1.0005	1
	UPS	0.8474	87	0.9057	12
	XOM	0.7926	36	1.0183	2
2002.07–12	IBM	0.8511	93	0.9452	48
	LEE	0.8419	79	0.9471	92
	MOT	0.8410	98	0.9348	97
	UPS	0.5995	1	0.9930	7
	XOM	0.8363	113	0.9069	11
2003.01–06	IBM	0.6293	3	0.9119	16
	LEE	0.8223	57	1.0113	3
	MOT	0.8636	84	0.9405	46
	UPS	0.8434	61	0.9443	61
	XOM	0.7554	21	0.9465	18
2003.07–12	IBM	0.5997	1	1.0003	1
	LEE	0.7470	28	0.9982	5
	MOT	0.6602	5	0.9821	8
	UPS	0.8189	54	0.9994	13
	XOM	0.7825	31	0.9945	31
2004.01–06	IBM	0.7531	20	0.9276	12
	LEE	0.6003	1	0.9726	7
	MOT	0.6072	1	0.9908	6
	UPS	0.8087	36	0.9872	25
	XOM	0.5997	1	0.9628	18
2004.07–12	IBM	0.8139	36	1.0075	3
	LEE	0.7464	16	0.9524	11
	MOT	0.8426	94	0.998	3
	UPS	0.8201	55	0.9935	55
	XOM	0.5996	1	1.0085	4
Average		0.7622	45.28	0.9685	22.2
Std. dev.		0.012		0.0015	

資金
調整
次數

Table 8

The result of paired t test on fitness values of CPPI and DPPI

H_0 :	$\mu_D - \mu_C \leq 0$
H_1 :	$\mu_D - \mu_C > 0$
D:	DPPI
C:	CPPI
Decision: Reject H_0 if $t > t_{(0.95,58)} = 1.6716$	
Result: $t = 10.8692 > 1.6716$, H_0 is rejected	

Table 9

PCA results of the leading risk variables for the risk multiplier

Variable	PCA value
T-bill	0.817
STO-D	0.709
NYSE	0.686
US2Euro	0.649
ST	0.544
%R	0.507
MT	0.446
STO-K	0.437
MFI	0.359

Table 10
PCA results by year

	2002	2003	2004
1	STO-D	T-bill	NYSE
2	T-bill	US2Euro	T-bill
3	US2Euro	MT	STO-D
4	ST	STO-D	%R
5	NYSE	MFI	STO-K
6	%R	NYSE	ST

Table 11
PCA results by company

	IBM	LEE	MOT	UPS	XOM
1	T-bill	T-bill	STO-D	ST	STO-D
2	US2Euro	STO-D	NYSE	MT	ST
3	NYSE	%R	US2Euro	T-bill	%R
4	MFI	ST	T-bill	%R	US2Euro
5	%R	NYSE	MT	NYSE	MFI
6	MT	US2Euro	STO-K	US2Euro	T-bill

Thank you!!

Disparity index

- a percentage display of the latest close to a chosen moving average

$$DI = ((C - EMA_n) / EMA_n) * 100$$

RSI : 判斷個股在某一段時間內超買超賣狀況的指數， range : 0~100

$$RSI = 100 - (100 / (1 + RS))$$

$$RS = AUn / ADn$$

AUn 表示 n 日內收盤價上漲點數的 n 日平均數 $= \Sigma(\text{上漲點數}_i) / n$

ADn 表示 n 日內收盤價下跌點數的 n 日平均數 $= \Sigma(\text{下跌點數}_i) / n$

Williams %R : 判斷個股在某一段時間內超買超賣狀況的指數， range : 0~100

$$\%R[n] = (H_n - C) / (H_n - L_n) * 100$$

其中 H_n 為 n 日內的最高價，

L_n 為 n 日內的最低價

C 為當日的收盤價。

- **Stochastic oscillator(STO)**, range : 0~100

$$\%K[n] = (1 - a) * \%K[n-1] + a * (C - L_n) / (H_n - L_n) * 100$$

$$\%D[n] = (1 - a) * \%D[n-1] + a * \%K[n]$$

其中 H_n 為 n 日內的最高價

L_n 為 n 日內的最低價

C 為當日的收盤價。

Back

Money flow index

Typical Price = (Highest + Lowest + Close) / 3

Raw Money Flow = Typical Price x Volume

Positive Money Flow =

Sum of Raw Money Flow for the specified number of periods where Typical Price increased

Negative Money Flow =

Sum of Raw Money Flow for the specified number of periods where Typical Price decreased

Money Ratio = Positive Money Flow / Negative Money Flow

Money Flow Index = $100 - (100 / (1 + \text{Money Ratio}))$

